Due: Sep 10, 11:59pm

1. Recall the spherical coordinates map from class:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} r\sin\theta\cos\varphi \\ r\sin\theta\sin\varphi \\ r\cos\theta \end{pmatrix}.$$

Let $f \in C^1(\mathbb{R}^3 \setminus 0)$ be a function of x. Compute $\partial_r f$, $\partial_\theta f$, and $\partial_\varphi f$ in terms of partial derivatives in x and vice versa. Compute the Laplacian $\Delta = \sum_{j=1}^3 \partial_{x_j}^2$ in spherical coordinates.

2. This problem explores matrix exponentials.

(a) Let $A, B \in \mathrm{Mat}(n \times n)$. Here $\mathrm{Mat}(n \times n)$ is the space of $n \times n$ matrices equipped with the operator norm

$$\|A\| := \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Ax\|}{\|x\|}$$

where $\|x\|^2 := \sum_j x_j^2$ is the usual Euclidean norm. Show that

$$\|AB\| \leq \|A\| \|B\|.$$

(b) Show that the series

$$e^A =: \exp(A) := \sum_{j=0}^{\infty} \frac{A^j}{j!}$$

converges in $\mathrm{Mat}(n \times n)$.

(c) Compute

$$\exp \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Hint: Diagonalize the matrix.

(d) Suppose A and B commute, that is $AB = BA$. Show that

$$e^A e^B = e^{A+B}.$$

(e) Give an example of $A, B \in \mathrm{Mat}(2 \times 2)$ such that

$$e^A e^B \neq e^{A+B}.$$

(f) (optional) Show that

$$\frac{d}{dt} e^{tA} = A e^{tA}$$

3. Find all functions $f : \mathbb{R} \to \mathbb{C}$ (function taking real numbers to complex numbers) such that

$$u(t, x) = f(x - ct)$$

solves

$$i\partial_t u + \frac{1}{2}\partial_x^2 u = 0.$$

This is the one-dimensional Schrödinger equation.

4. Some practice with point-set topology:

   (a) We defined in class that $F \subset \mathbb{R}^n$ is *closed* if the complement F^C is open. Show that F is closed if and only if every sequential limit of F lies in F. Here, $y \in \mathbb{R}^n$ is a *sequential limit* of F if there exists a sequence $(x_j)_{=1}^{\infty}$ in F such that $\lim_{j\to\infty} x_j = y$.

   (b) We defined in class that $K \subset \mathbb{R}^n$ is *compact* if it is closed and bounded. Consider the simpler case $K \subset \mathbb{R}$. Show that K is compact if and only if every sequence in K has a convergent subsequence with limit in K.

   Comment 1: Do not cite Bolzano–Weierstrass. The purpose of this exercise is to repeat and absorb the proof for yourself.

   Comment 2: Think about how this proof can be generalized to $\mathbb{R}^n$. How-ever, it is crucial that that we are working in *inite* dimensional spaces. The equivalence fails for infinite dimensional spaces such as functional spaces.

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